Artificial Intelligence 700

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Qualification: BSC **Year of Study:** 2026 **Semester:** 01

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS
QUESTION 1			
QUESTION 2			

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document.

Ensure you show ALL your steps and clearly outline your solution for all questions

Submit the assignment in PDF format on Moodle.
Use the specified cover page provided.
Include a signed declaration of originality.

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I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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QUESTION 1

1.1 **Defining AI: The four Approaches framework**.

a) The Turing Test, which was proposed by Alan Turing, is a method used to determine whether a machine can exhibit Intelligent behaviour indistinguishable from that of a human. A machine passes the test if a human evaluator cannot reliably differentiate between the machine and a human.

Capabilities a machine must demonstrate to pass the Turing test:

- Natural Language processing(NLP): to understand and generate human-like language.
- Machine learning: to adapt and improve from experience.

Examples:

1. Engaging in a natural conversation with a human.
2. Solving logical or general knowledge questions.

b) **Perception:**

The car detects the child 30 meters ahead using sensors and identifies speed which is 50km/h and surroundings.

Reasoning:

The system calculates stopping distance, evaluates risks and considers possible actions such as braking or swerving.

Action:

The car immediately brakes and if necessary, swerves safely to avoid collision.

Rational behavior:

The system prioritizes human safety and selects the action that minimizes harm.

c) The Thinking Humanly approach focuses on stimulating human thought processes(cognitive modelling), while the Thinking Rationally approach uses logical rules to make optimal decisions.

Thinking Humanly is difficult to implement accurately due to the complexity of human cognition. Thinking Rationally is more useful for building practical AI systems today, as it focuses on correct and efficient decision making.

Examples:

1. Search engines use algorithms to produce optimal results.
2. Self driving cars use logical decision making rather than human-like thinking.

1.2 Bias in AI systems: Sources and manifestations

a) The types of Bias

- Data Bias – Amazon Recruitment

The system was trained on historical hiring data that favored male candidates, causing it to learn and replicate this bias by penalizing resumes associated with women.

- Algorithmic Bias – Compas System

The model produced biased predictions due to how the algorithm processed and weighed features, resulting in unfairly higher risk scores for black defendants.

- Facial Recognition Systems – Deployment Bias

The systems performed poorly in real-world settings due to insufficient testing on diverse populations, leading to higher error rates for certain groups.

b) Machine learning models learn patterns from historical data. If the data reflects past discrimination, the model identifies these patterns as valid and reproduces them in its predictions leading to Bias amplification, where existing inequalities are reinforced.

Training on historical data does not guarantee fairness because the data may already contain societal biases and the model optimizes accuracy rather than fairness.

c) **Technical interventions**

1. Fairness Constraints: Modify the model to ensure equal prediction outcomes across different groups.
2. Bias detection tools: Evaluate model outputs using fairness metric(e.g false positive across groups).

Organizational/policy interventions:

1. Diverse training data: Ensure datasets include balanced representation of all demographic groups.
2. Regular audits: Conduct independent reviews of AI systems to identify and correct bias before deployment.

1.3 AI Transparency, explainability and Accountability

a) The black box problem refers to the lack of transparency in AI systems, where the internal decision making process is not easily interpretable by humans. While the system produces outputs, it does not clearly explain how those outputs were deprived.

Deep neutral networks are particularly difficult to interpret because they consist of many hidden layers with millions of parameters. These layers transform input data in complex, non-linear ways that are not easily traceable.

On the other hand, traditional systems such as rule based models and decision trees are more interpretable because they follow explicit rules or structured paths that can easily be understood and traced by humans.

b) Explainability Techniques

1. Lime(Local Interpretable Model-Agnostic Explanations)

- Provides local explanations by approximating the model around a specific prediction.
- Helps identify which features influenced the decision.
- Limitation: It only explains individual predictions, not the entire model.

2. SHAP(Shapely Additive Explanations)

- Assigns importance values to each feature based on its contribution to the prediction.
- Offers consistent and theoretically grounded explanations.
- Limitation: Computationally expensive for complex models

3. Saliency Maps

- Highlights important regions in the X-ray image that influenced the prediction.
- Helps radiologists visually understand what the model
- Limitation: May not fully explain the reasoning process behind the prediction.

c) Accountability should be viewed as a shared responsibility among multiple stakeholders.

- AI Developers: Responsible for designing, training and validating the system.
- Hospital Administrators: Responsible for approving and deploying the AI system.
- Radiologist: Responsible for reviewing and making the final clinical decision

From legal perspective, liability may be shared depending on system failure or misuse.

From ethical perspective, human oversight is essential especially in high risk domains like healthcare.

From a practice perspective, the AI acts as a decision support tool, meaning the radiologist retains final responsibility for patient care.

QUESTION 2

2.1 Reinforcement Learning: Grid World

MDP Formulation

The problem can be modelled as a Markov Decision Process defined as follows:

- State space(S): All possible grid positions(0,0) to (3,3)
- Action space(A): {Up, Down, Left, Right}
- Transition function P(s′|s,a): Deterministic movement where each action leads to a new state unless blocked by boundaries.
- Reward function $R(s, a, s')$: +1 for reaching the goal, -1 for hitting an obstacle and -0.01 per step
- Discount factor(y): 0.95

The Markov property holds because the next state depends only on the current state and action, not on past states

Exploration vs Exploitation

The agent must balance:

- Exploration: trying new actions to discover better paths
- Exploitation: choosing actions with the highest known reward

Epsilon-greedy strategy(ε = 0.3):

- 30% probability → choose a random action
- 70% probability → choose the best action

At state (0,1):

Q-values: Right = 0.5, Down = 0.2, Up = 0, Left = 0

→ Best action = Right(0.5)

→ Probability of choosing Right = 0.7 + (0.3 ÷ 4) = 0

→ Other actions = 0.075 each

Q-Learning Update Rule

Formular: Q(s,a)=Q(s,a)+α[r+γmaxQ(s′,a′)−Q(s,a)]

Where:

- : current value
- α: learning rate
- r: reward
- γ: discount factor
- max $Q(s', a')$: best future value

Calculation: Given

Q = 0.3, α = 0.1, r = −0.01, γ = 0.95, terminal → max Q = 0

$$Q = 0.3 + 0.1[(-0.01 + 0) - 0.3]$$
$$Q = 0.3 + 0.1(-0.31) = 0.269$$

Optimal path Explanation

The path
(0,0) → (0,1) → (0,2) → (0,3) → (1,3) → (2,3) → (3,3)
Is optimal because it:

- avoids obstacles (-1 penalty)
- minimizes steps (-0.01 cost per step)

if the step penalty is removed:

- All paths equal steps become similar in value
- The agent may explore longer or inefficient routes

Q-Learning vs Deep Q-Networks(DQN)

- Q-Learning: Uses a table to store Q-values → suitable for small environments
- DQN: Uses neutral networks → suitable for large complex environments(e.g Atari games)

Why DQN is needed:

- Cannot store all states in large environments
- Handles high-dimensional inputs(e.g images)

Two key innovations:

1. Experience replay: Stores past experiences to improve learning stability
2. Target network: Uses a separate network to stabilize updates

2.2 **Word Embedding and Language Modelling**
Static vs Contextual Embeddings

Word2Vec(Static Embeddings):
Word2Vec assigns a single fixed vector to each word regardless of context. This means the meaning of a word does not change across different sentences.

BERT(Contextual Embeddings)
BERT generates context dependent embeddings, where the meaning of a word changes based on surrounding words in a sentence.

Example using bank:

- River bank → Word2Vec: Same vector; BERT: embedding reflects “edge of river”

- Savings bank → Word2Vec: same vector; BERT: embedding reflects "financial institution"

BERT handles polysemy(multiple meanings) more effectively than Word2Vec.

Bigram probabilities(MLE)

Corpus: "The doctor prescribed medicine. The medicine worked well."

1. P(medicine | The)

$$P(\text{medicine} \mid \text{The}) = \frac{\text{Count(The medicine)}}{\text{Count(The)}}$$

- Count(The medicine) = 1
- Count(The) = 2

$$P = \frac{1}{2} = 0.5$$

2. P(worked | medicine)

$$P(\text{worked} \mid \text{medicine}) = \frac{\text{Count(medicine worked)}}{\text{Count(medicine)}}$$

- Count(medicine worked) = 1
- Count(medicine) = 2

$$P = \frac{1}{2} = 0.5$$

<u>Advantages of Transformers over N-gram models</u>

1. Context awareness:

Transformers consider the full sentence context, improving understanding of complex medical queries.

2. Long-range dependency handling:

They capture relationships between distant words, which is essential for interpreting medical descriptions.

3. Higher accuracy and adaptability:

Models like BERT are pre-trained on large datasets and fine-tuned for tasks such as diagnosis support in chatbots.

2.3 NPL Preprocessing Pipeline with code.

```
import nltk
From nltk.tokenize import word_tokenize
From nltk.corpus import stopwords
From nltk.stem import wordNetLemmatizer

review1 = "The AI chatbot's responses were ABSOLUTELY AMAZING!!! It understood my queries perfectly…"
```

review2 = “Terrible experience. The NLP model couldn’t understand simple questions….”

Stop_words = set(stopwords.words(`english`))
Lemmatizer = wordNetLemmatizer()

Def preprocess(text):
tokens = word_tokenize(text)
tokens = [word.lower() for word in tokens]
tokens = [word for word in tokens if word.isalpha()]
tokens = [lemmatizer.lemmatize(word) for word in tokens]
Returns tokens
Print(preprocess(review1))
Print(preprocess(review2))

Preprocessed Output (Example)

Review 1:
[`ai`, `chatbot`, `response`, `absolutely`, `amazing`, `understood`, `query`, `perfectly`]
Review 2:
[`terrible`, `experience`, `nlp`, `understand`, `simple`, `question`]

Importance of Each Step

- Tokenisation: Splits text into words for analysis
- Lowercasing: Ensures uniformity(e.g “Amazing” = “amazing”)
- Stop-word removal: Eliminates common words that add little meaning
- Lemmatization: Reduces words to base forms(e.g “running” → “run”)

These steps improve performance in tasks such as sentiment analysis and text classification.

Issues with Aggressive preprocessing

1. Loss of sentiment intensity
Removing punctuation e.g”!!!” reduces emotional emphasis, affecting sentiment detection.
2. Loss of meaning in negation:
Removing words like “not” can change meaning e.g “not good” → “good” , leading to incorrect classification.

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